Ivan Bercovich
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Overview

I am experienced in leading engineering/product teams in the 100–200 person range. I also have general management experience at Amazon, where I opened and led the office in Santa Barbara, and as a CEO with HeyTutor. I led the 150-person technical team that was acquired by Alexa as a pre-LLM technology to power its NLP and Q&A capabilities. After Amazon, I started a venture capital firm, ScOp Venture Capital, which currently has ˜$200M AUM. I've invested in the first rounds of vertical AI companies such as Rogo.ai, ChipAgents.ai, Pangram, and Harbor. Over the past couple of years, I've been interested in AI Safety. To stay grounded in the research, I lead a team focused on Reward Hacking (github.com/few-sh/terminal-wrench), and I am a core contributor to Terminal Bench, a benchmark focused on measuring SOTA model capabilities that is regularly included in model cards.

Experience

Independent Researcher — July 2025 - Present
After attending The Curve in 2024, I became deeply concerned with AI Safety. I shifted my energy towards intersecting topics such as benchmarking, in collaboration with Terminal Bench, and reward hacking (supported by a $870k grant from Coefficient) on Terminal Bench and Kernel Bench. I'm currently very interested in the process of developing great benchmarks, such as what makes a task truly difficult.

Founder/Partner: ScOp Venture Capital — 2021 - Present
I founded ScOp with 3 partners to leverage my operating experience across a roughly $200M portfolio. Depending on the needs of the founders, I might act as fractional CTO, sit on the board, or provide ad-hoc advice. Among my partners, I focus primarily on technical discovery and diligence, and general operating support (product/tech strategy, recruiting, org building, etc).

Investor/CEO: HeyTutor — 2022 - 2023
While working as an investor, I joined as interim CEO for a year to help HeyTutor's transition from startup to growth business. HeyTutor works with school districts across the country to deliver supplemental instruction. We adopted stricter procedures and executed meaningful organizational changes for a workforce of 1000+ employees and tutors. We grew the business by 5x, to $36M in revenue, and became profitable.

Investor/Founder: Unwrap.ai — 2020 - 2021
I was the co-founder and first CEO of Unwrap, a startup I incubated with the Allen Institute for AI. We focused on making sense of customer feedback at scale. I built the initial models, hired the first engineer, and raised seed capital, but ultimately decided to pursue investing full time.

GM Knowledge Graph: Amazon Alexa — 2017 - 2020
Amazon acquired our company, Graphiq, to improve Alexa's open domain question understanding and answering (Q&A) capabilities. At the time, 2017, Alexa was behind competitors in the Q&A space, but assisted by the Graphiq acquisition, Alexa caught up*[1] to be near or above the best-in-class assistant in most knowledge categories. When Graphiq's CEO left, I became the GM for the team of 150 software developers and knowledge engineers (KE), who jointly built the technology and knowledge graph (KG) supporting Alexa's semantic Q&A capabilities. By the time I ended my tenure in 2020, the Graphiq technology was handling most Q&A across all languages and locales in which Alexa was present. We developed solutions for natural language processing, knowledge retrieval, and language generation. In parallel, we built a number of internal tools and workflows that enabled our KEs to define complex knowledge representations and ingest billions of facts to build one of the most comprehensive KGs.

VP of Engineering: Graphiq — 2011 - 2017
I started as a software developer, then managed the 40-person engineering team as VP, and eventually succeeded the CEO. Graphiq focused on collecting, structuring, and connecting the world's data — billions of entities and hundreds of billions of facts. We began by building stand-alone vertical search engines for specific categories, but upon understanding the value of interconnected information, we progressively developed a complex ontology and KG. Our website and embeddable widgets allowed people to thoroughly

[1] *Amazon 2018 shareholder letter: "Last year, we improved Alexa's ability to understand requests and answer questions by more than 20%, while adding billions of facts to make Alexa more knowledgeable than ever."

research thousands of topics on one intuitive interface. Launched in late 2010, Graphiq quickly became a leading research engine with 40+ million monthly visits, and 400 million monthly impressions of our KG across the internet. In 2016, as voice assistants became better known, we invested in novel approaches using natural language processing for KG retrieval, leading to multiple acquisition offers.

Cisco Systems — 2009
I worked as a UX engineer on Cisco Quad - an attempt to build a social network for the enterprise.

Education

University of California, Santa Barbara (unfinished) — 2010
Ph.D. in Statistics and Applied Probability with emphasis in Financial Mathematics.

University of Massachusetts Amherst — 2005 - 2009
BS in Electrical Engineering and Mathematics, Summa Cum Laude. GPA 3.95/4.00
Departmental Honors with greatest distinction and top 2%.

Research

Selected work since shifting my focus to AI Safety, centered on benchmarking and reward hacking.

Publications

2025

- **Agents of change: Self-evolving LLM agents for strategic planning**. N Belle, D Barnes, A Amayuelas, I Bercovich, XE Wang, W Wang. *arXiv:2506.04651.*
- **Hardtests: Synthesizing high-quality test cases for LLM coding**. Z He, YM Choi, K Zhang, J Ji, J Zhou, D Xu, I Bercovich, A Zhang, L Li. *arXiv:2505.24098.*
- **MessIRve: A large-scale Spanish information retrieval dataset**. F Valentini, V Cotik, D Furman, I Bercovich, E Altszyler, JM Pérez. *EMNLP 2025.*

2026

- **Terminal Wrench: A Dataset of 331 Reward-Hackable Environments and 3,632 Exploit Trajectories**. I Bercovich, I Segal, K Zhang, S Saxena, A Raghunathan, Z Zhong. *arXiv:2604.17596.*
- **Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces**. MA Merrill, AG Shaw, N Carlini, B Li, H Raj, I Bercovich, L Shi, JY Shin, et al. *arXiv:2601.11868.*

Under Review

Three papers under review at NeurIPS 2026.

- **Hardening Agent Benchmarks with Adversarial Hacker-Fixer Loops**. Automatically building exploit-resistant verifiers for agent benchmarks using adversarial loops of cooperating LLM agents.
- **SWE-Marathon: Long-Horizon Software Engineering Benchmark**. Project-scale software tasks spanning hours and millions of tokens, designed to resist single-test shortcut solutions.
- **Search strategies for autonomous ML research agents**. How search and quality-diversity strategies affect autonomous agents' ability to explore and produce novel machine learning research.

In Progress

Training secure coding agents. A follow-up to SusVibes, moving from benchmarking to training. The original work showed that frontier models produce functionally correct but insecure code. This project improves data synthesis methods and explores training outcomes.

Training coding agents with privileged information. Coding agents learn from sparse, binary rewards, so most rollouts yield no training signal. This project injects oracle hints (such as the fix patch) at training time to elicit correct trajectories the model can't reach unaided, and uses action-level entropy to decide when a teacher should guide the agent versus let it act on its own.